



KTU NOTES

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Module V : STATE-SPACE ANALYSIS

- Modern control theory is based on state-space approach which is used for the analyse and design of control systems.
- It can be applied to linear and non-linear, time-variant and time-invariant, SISO and MIMO systems.

SISO — Single Input Single Output

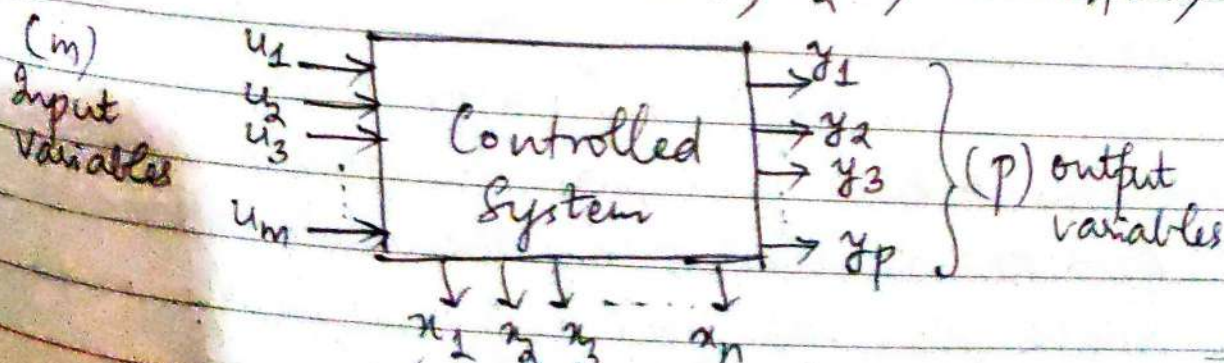
MIMO — Multi Input Multi Output

- In state-space analyse, a system is represented by a mathematical model consisting of a set of input output and state variables related by first-order differential equations.

Input variables : $u_1(t), u_2(t), \dots, u_m(t)$

Output variables : $y_1(t), y_2(t), \dots, y_p(t)$

State variables : $x_1(t), x_2(t), \dots, x_n(t)$



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Input, output and state variables can be represented in vector form as:

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix} \quad (m \times 1) \quad ; \quad y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix} \quad (p \times 1) \quad \text{and} \quad x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \quad (n \times 1)$$

For a general system, the state-space representation can be arranged in the form of n first-order differential equations:

$$\left. \begin{aligned} \frac{dx_1}{dt} = \dot{x}_1 &= f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m) \\ \frac{dx_2}{dt} = \dot{x}_2 &= f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m) \\ &\vdots \\ \frac{dx_n}{dt} = \dot{x}_n &= f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m) \end{aligned} \right\} \text{--- (1)}$$

Equation (1) can be written in vector form as:

$$\dot{x}(t) = f(x(t), u(t)) \quad \text{--- (2)}$$

27 SUNDAY The output $y(t)$ can in general be expressed in terms of the state $x(t)$ and input $u(t)$ as $y(t) = g(x(t), u(t))$ --- (3)

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Equation (2) is called the state equation and equation (3) is called the output equation for time-invariant systems.

For time-varying systems, the corresponding eqns are:

$$\dot{x}(t) = f(x(t), u(t), t)$$

$$y(t) = g(x(t), u(t), t)$$

Terms used in state-space analysis:-

(1) State:- The state of a dynamical system is a minimal set of variables such that the knowledge of these variables at $t = t_0$ together with the knowledge of the inputs for $t \geq t_0$ completely determines the behaviour of the system for $t \geq t_0$.

(2) State variables:- The minimal set of variables which determine the dynamics of a linear system are called state variables. An n^{th} -order system has 'n' state variables $x_1(t), x_2(t) \dots x_n(t)$.

(3) State vector:- The n state variables are the components of a vector $x(t)$ called state vector.

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$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

(4) State-space:- The n -dimensional space whose coordinate axes consist of x_1 -axis, x_2 -axis, ..., x_n -axis is called a state-space.

Comparison of classical (conventional) control theory and modern control theory

<u>Classical control theory</u>	<u>Modern control theory</u>
1) Frequency domain approach	1) Time domain approach
2) Based on transfer function	2) Based on state-space analysis
3) Can be applied only to LTI, SISO systems	3) Can be applied to SISO and MIMO systems, ATV and Linear and Nonlinear systems, Time-Varying and Time invariant systems.

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4) Initial conditions are assumed to be zero

4) Initial conditions are taken into account

5) Does not provide information regarding the internal states of the system

5) Provides information regarding the internal states of the system

6) Design is carried out for specific inputs such as impulse, step, ramp, parabolic etc

6) Design can be carried out for a class of inputs

7) Classical design methods based on T.F model are essentially trial and error procedures which may not lead to optimal control systems

7) Optimal control systems can be designed easily for the given performance indices

8) It is difficult to handle large-scale complex systems

8) It is ~~not~~ easy to handle large-scale complex systems with the help of digital computers

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STATE MODEL OF LINEAR SYSTEMS

The state and output equations constitute the state model of a system.

$$\dot{x}(t) = Ax(t) + Bu(t) \quad \text{--- state equation}$$

where $x(t)$ is $n \times 1$ state vector

$u(t)$ is $m \times 1$ input vector

A is $n \times n$ system matrix

defined by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

B is $n \times m$ input matrix defined by

$$B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix}$$

$$y(t) = Cx(t) + Du(t) \quad \text{--- output equation}$$

where $y(t)$ is $p \times 1$ output vector

C is $p \times n$ output matrix

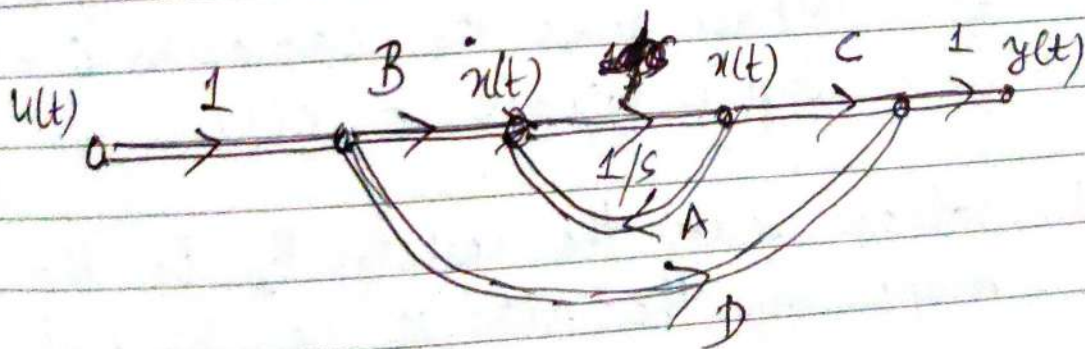
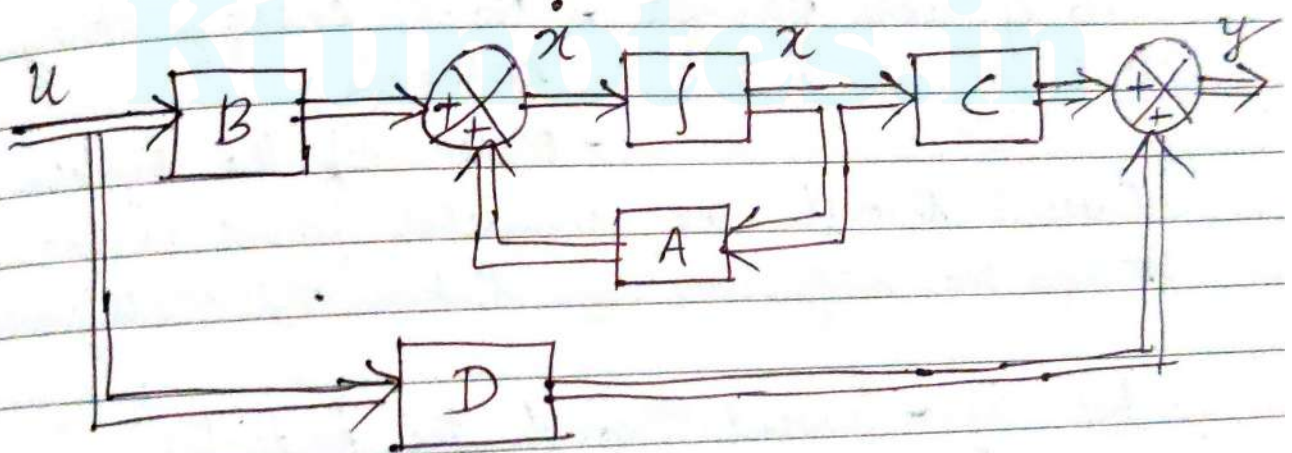
defined by

$$C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{p1} & c_{p2} & \dots & c_{pn} \end{bmatrix}$$

D is $p \times m$ transmission matrix defined by

$$D = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1m} \\ d_{21} & d_{22} & \dots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{p1} & d_{p2} & \dots & d_{pm} \end{bmatrix}$$

State diagram: - The pictorial representation of the state model of a system either in block diagram form or in signal flow graph form is called state diagram.



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Derivation of T.F from state Model

For a general state model,

$$\dot{x} = Ax + Bu \quad \text{--- state equation}$$

$$y = Cx + Du \quad \text{--- output equation}$$

Taking Laplace transforms of the above:-

$$sX(s) - x(0) = AX(s) + BU(s) \quad \text{--- (1)}$$

$$Y(s) = CX(s) + DU(s) \quad \text{--- (2)}$$

T.F assumes zero initial conditions

$$\therefore (3) \Rightarrow sX(s) = AX(s) + BU(s)$$

$$\Rightarrow [sI - A]X(s) = BU(s)$$

$$\Rightarrow X(s) = [sI - A]^{-1} BU(s) \quad \text{--- (3)}$$

Substituting (3) in (2) :-

$$Y(s) = C[sI - A]^{-1} BU(s) + DU(s)$$

$$\boxed{TF = \frac{Y(s)}{U(s)} = C[sI - A]^{-1} B + D}$$

1) For the state models given below, determine the transfer function.

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$$a) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T.F = C [sI - A]^{-1} B + D$$

$$\text{Where } [sI - A]^{-1} = \frac{\text{Adj}[sI - A]}{|sI - A|}$$

$$[sI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\text{Adj}[sI - A] = \text{Transpose of cofactor matrix of } [sI - A]$$

$$\text{Cofactor matrix of } [sI - A] = \begin{bmatrix} s+3 & -2 \\ -1 & s \end{bmatrix}$$

$$\text{Adjoint of } [sI - A] = \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$|sI - A| = \frac{s(s+3)+2}{s^2+3s+2}$$

$$\therefore [sI - A]^{-1} = \frac{\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}}{s^2+3s+2}$$

$$T.F = C [sI - A]^{-1} B + D$$

$$= \frac{\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}}{s^2+3s+2}$$

$$= \frac{1}{s^2+3s+2} \begin{bmatrix} -2 & s \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{-2}{s^2+3s+2}$$

$$b) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 40 & 10 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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$$[sI-A] = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 3 & s+4 \end{bmatrix}$$

$$[sI-A]^{-1} = \frac{\text{Adj}[sI-A]}{\det[sI-A]}$$

cofactor matrix of $[sI-A]$

$$= \begin{bmatrix} s(s+4)+3 & 0 & 0 \\ s+4 & s(s+4) & -3s \\ 1 & s & s^2 \end{bmatrix}$$

$$\text{Adj}[sI-A] = \begin{bmatrix} s^2+4s+3 & s+4 & 1 \\ 0 & s^2+4s & s \\ 0 & -3s & s^2 \end{bmatrix}$$

$$|sI-A| = s(s^2+4s+3)$$

$$= s^3+4s^2+3s$$

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$$[sI - A]^{-1} = \frac{\text{Adj}[sI - A]}{|sI - A|}$$

$$= \frac{1}{s^3 + 4s^2 + 3s} \begin{bmatrix} s^2 + 4s + 3 & s + 4 & 1 \\ 0 & s^2 + 4s & s \\ 0 & -3s & s^2 \end{bmatrix}$$

$$T.F = C[sI - A]^{-1}B + D$$

$$= \begin{bmatrix} 40 & 10 & 0 \end{bmatrix} \begin{bmatrix} s^2 + 4s + 3 & s + 4 & 1 \\ 0 & s^2 + 4s & s \\ 0 & -3s & s^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$s^3 + 4s^2 + 3s$$

$$= \begin{bmatrix} 40(s^2 + 4s + 3) & 40(s + 4) + 10(s^2 + 4s) & 40 + 10s \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$s^3 + 4s^2 + 3s$$

$$= \frac{10s + 40}{s^3 + 4s^2 + 3s}$$

SOLUTION OF STATE EQUATIONS

1) Homogeneous case (Unforced system): -

Consider the first order scalar differential equation,

$$\frac{dx}{dt} = ax \quad ; \quad x(0) = x_0$$

This equation has the solution

$$x(t) = e^{at} x_0 \\ = \left[1 + at + \frac{a^2 t^2}{2!} + \dots \right] x_0$$

Now consider the state equation

$$\dot{x}(t) = A x(t) \quad ; \quad x(0) = x_0 \quad - (1)$$

which represents a homogeneous linear system (unforced system) with constant coefficients.

By analogy with the scalar case, we assume a solution of the form

$$x(t) = a_0 + a_1 t + a_2 t^2 + \dots \quad - (2)$$

where a_i 's are vector coefficients

By substituting the assumed solution of (2) in (1), we get: -

$$a_1 + 2a_2 t + \dots = A(a_0 + a_1 t + a_2 t^2 + \dots)$$

Comparing coefficients: -

$$a_1 = A a_0 \\ a_2 = \frac{A a_1}{2} = \frac{A^2 a_0}{2!}$$

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In the assumed solution, put $t=0$

$$\text{Then } x(0) = x_0 = a_0 \quad \text{--- (3)}$$

The solution $x(t)$ is thus

$$x(t) = a_0 + A a_0 t + \frac{A^2 a_0 t^2}{2!} + \dots$$

$$= \left[I + At + \frac{A^2 t^2}{2!} + \dots \right] a_0$$

$$= \left[I + At + \frac{A^2 t^2}{2!} + \dots \right] x_0$$

($\because a_0 = x_0$ from (3))

$$\text{Define } I + At + \frac{A^2 t^2}{2!} + \dots = e^{At}$$

which is called a matrix exponential

Thus, the solution $x(t)$ can now be written as

$$x(t) = e^{At} x_0 \quad \text{--- (4)}$$

From equation (4), it is observed that the initial state x_0 at $t=0$ is driven to a state $x(t)$ at time t . This transition in state is carried out by the matrix exponential e^{At} . Hence it is also called as STATE TRANSITION MATRIX denoted by $\phi(t)$.

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(ii) Non-homogeneous case (Forced system):-

$$\dot{x}(t) = Ax(t) + Bu(t) ; x(0) = x_0$$

$$\Rightarrow \dot{x}(t) - Ax(t) = Bu(t)$$

Multiplying both sides by e^{-At}

$$e^{-At} [\dot{x}(t) - Ax(t)] = e^{-At} Bu(t)$$

$$\Rightarrow \frac{d}{dt} (e^{-At} x(t)) = e^{-At} Bu(t)$$

Integrating both sides w.r.t t between the limits 0 and t , we get

$$e^{-At} x(t) \Big|_0^t = \int_0^t e^{-A\tau} Bu(\tau) d\tau$$

$$e^{-At} x(t) - x(0) = \int_0^t e^{-A\tau} Bu(\tau) d\tau$$

Pre-multiplying both sides by e^{At}

$$e^{At} [e^{-At} x(t) - x(0)] = e^{At} \int_0^t e^{-A\tau} Bu(\tau) d\tau$$

$$\Rightarrow x(t) = \underbrace{e^{At} x(0)}_{\text{Homogeneous solution}} + \underbrace{\int_0^t e^{A(t-\tau)} Bu(\tau) d\tau}_{\text{Forced response}}$$

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Properties of STM ($STM = \phi(t) = e^{At}$)

$$1) \phi(0) = I$$

$$\phi(0) = e^{A \times 0} = I$$

$$2) \phi^{-1}(t) = \phi(-t)$$

Proof:-

$$\phi^{-1}(t) = (e^{At})^{-1} = e^{-At} = \phi(-t)$$

$$3) \phi(nt) = [\phi(t)]^n$$

Proof:-

$$\begin{aligned} \phi(nt) &= e^{Ant} \\ &= (e^{At})^n \\ &= [\phi(t)]^n \end{aligned}$$

$$4) \phi(t_1 + t_2) = \phi(t_1) \phi(t_2) = \phi(t_2) \phi(t_1)$$

$$LHS = \phi(t_1 + t_2)$$

$$= e^{A(t_1 + t_2)} = e^{At_1} \cdot e^{At_2} \quad \text{--- (1)}$$

$$RHS = \phi(t_1) \phi(t_2)$$

$$= e^{At_1} e^{At_2} \quad \text{--- (2)}$$

$$\text{Thus } \phi(t_1 + t_2) = \phi(t_1) \phi(t_2) = \phi(t_2) \phi(t_1)$$

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$$5) \phi(t_2 - t_1) \cdot \phi(t_1 - t_0) = \phi(t_2 - t_0)$$

$$\phi(t_2 - t_1) \cdot \phi(t_1 - t_0) = e^{A(t_2 - t_1)} \cdot e^{A(t_1 - t_0)}$$

$$= e^{A(t_2 - t_1 + t_1 - t_0)}$$

$$= e^{A(t_2 - t_0)}$$

$$= \phi(t_2 - t_0)$$

Computation of STM (by Laplace transformation)

(i) $\dot{x} = Ax$ (unforced system)
Taking Laplace transform

$$sX(s) - x(0) = AX(s)$$

$$[sI - A]X(s) = x(0)$$

$$\Rightarrow X(s) = [sI - A]^{-1} x(0)$$

Taking the inverse Laplace transform,

$$x(t) = L^{-1} [sI - A]^{-1} x(0)$$

$$\text{Thus, STM} = \phi(t) = e^{At} = L^{-1} [sI - A]^{-1}$$

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12	13	14	15	16	17	18	19	20	21	22
23	24	25	26	27	28	29	30	31		

$$\dot{x} = Ax + Bu \quad (\text{forced system})$$

(10) Taking h.T :-

Taking Laplace transform on both sides

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$[sI - A]X(s) = x(0) + Bu(s)$$

$$\Rightarrow X(s) = [sI - A]^{-1} x_0 + [sI - A]^{-1} B u(s)$$

Taking inverse L.T

$$x(t) = L^{-1}[(sI - A)^{-1}]x_0 + L^{-1}[(sI - A)^{-1} B U(s)]$$

$$= \phi(t)x_0 + L^{-1} \left[\phi(s) B u(s) \right]$$

Here $\phi(s) = (sI - A)^{-1}$ is called the resolvent matrix.

4 (1) Consider a control system with the state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u; \quad x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

(a) compute STM

(b) Compute the solution of state equation for a unit-step input

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SOLUTION:-

$$(1) \text{ STM} = \phi(t) = L^{-1} [SI - A]^{-1}$$

$$[SI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$[SI - A]^{-1} = \frac{\text{Adj}^o (SI - A)}{|SI - A|}$$

$$= \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

To find $L^{-1} [SI - A]^{-1}$:-

$$(1) \Rightarrow \frac{(s+3)}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$s+3 = A(s+2) + B(s+1)$$

$$s = -1 \Rightarrow 2 = A$$

$$s = -2 \Rightarrow 1 = -B$$

$$\Rightarrow \frac{2}{(s+1)} + \frac{-1}{(s+2)}$$

$$(2) \frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$1 = A(s+2) + B(s+1)$$

$$s = -1 \Rightarrow A = 1$$

$$s = -2 \Rightarrow B = -1$$

$$\Rightarrow \frac{1}{(s+1)} + \frac{-1}{(s+2)}$$

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$$\textcircled{3} \quad \frac{-2}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$-2 = A(s+2) + B(s+1)$$

$$s = -1 \Rightarrow -2 = A \quad \therefore A = -2$$

$$s = -2 \Rightarrow -2 = -B \quad B = 2$$

$$\Rightarrow \frac{-2}{s+1} + \frac{2}{s+2}$$

$$\textcircled{4} \quad \frac{s}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$+s = A(s+2) + B(s+1)$$

$$s = -1 \Rightarrow -1 = A \quad \therefore A = -1$$

$$s = -2 \Rightarrow -2 = -B \quad \therefore B = 2$$

$$\Rightarrow \frac{-1}{s+1} + \frac{2}{s+2}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{2}{s+2} + \frac{-1}{s+2} & \frac{1}{s+1} + \frac{-1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$$

$$p_{TM} = L^{-1} [sI - A]^{-1} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

2020

$$(ii) \quad x(t) = \underbrace{\phi(t)}_{\text{I}} x(0) + \underbrace{L^{-1} [sI - A]^{-1} B U(s)}_{\text{II}}$$

$$\text{I} \Rightarrow \phi(t) x(0) = \begin{bmatrix} e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$$

$$\text{II} \quad [sI - A]^{-1} B U(s) = \begin{bmatrix} \frac{2}{s+1} + \frac{-1}{s+2} & \frac{1}{s+1} + \frac{-1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$\times \frac{1}{s}$

$$\text{II} \quad [sI - A]^{-1} B U(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \frac{1}{s}$$

$$= \frac{1}{s(s^2 + 3s + 2)} \begin{bmatrix} 2 \\ 2s \end{bmatrix}$$

24 OCTOBER

	M	T	W	T	F	S	S	M	T	W	T	F	S	S
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
OCT	15	16	17	18	19	20	21	22	23	24	25	26	27	28
NOV	29	30	1	2	3	4	5	6	7	8	9	10	11	12

$$\frac{2}{s(s^2+3s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$2 = A(s+1)(s+2) + Bs(s+2) + Cs(s+1)$$

$$A = 1$$

$$B = -2$$

$$C = 1$$

$$\Rightarrow \frac{1}{s} + \frac{-2}{s+1} + \frac{1}{s+2}$$

$$\frac{2s}{s(s^2+3s+2)} = \frac{2}{s^2+3s+2} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$2 = A(s+2) + B(s+1)$$

$$s = -1 \Rightarrow 2 = A$$

$$s = -2 \Rightarrow 2 = -B \Rightarrow \frac{2}{s+1} + \frac{-2}{s+2}$$

$$\mathcal{L}^{-1} \left\{ [sI - A]^{-1} B U(s) \right\} = \mathcal{L}^{-1} \left[\begin{array}{c} \frac{1}{s} + \frac{-2}{s+1} + \frac{1}{s+2} \\ \frac{2}{s+1} + \frac{-2}{s+2} \end{array} \right]$$

$$= \begin{bmatrix} 1 - 2e^{-t} + e^{-2t} \\ 2e^{-t} - 2e^{-2t} \end{bmatrix}$$

25 SUNDAY

$$\text{Thus } x(t) = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix} + \begin{bmatrix} 1 - 2e^{-t} + e^{-2t} \\ 2e^{-t} - 2e^{-2t} \end{bmatrix}$$

2020

M	T	W	T	F	S	S	M	T	W	T	F	S	S
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23	24	25	26	27	28	29	30						

$$\Rightarrow x(t) = \begin{bmatrix} 1 - e^{-t} \\ e^{-t} \end{bmatrix}$$

2) The following facts are known about the linear system $\dot{x}(t) = Ax(t)$

if $x(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, then $x(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$

if $x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, then $x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$

Find e^{At} and hence A .

SOLUTION:-

For a homogeneous system,

$$x(t) = e^{At} x(0) \\ = \phi(t) x(0)$$

$$\Rightarrow \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

and $\begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Solving the above equations,

$$e^{At} = \phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & -e^{-t} + e^{-2t} \\ 2e^{-2t} - 2e^{-t} & e^{-t} - e^{-2t} \end{bmatrix}$$

27

OCTOBER

M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11			
12	13	14	15	16	17	18	19	20	21	22	23	24	25
26	27	28	29	30	31								

WK 44 • 2019
TUESDAYTo find A

$$\frac{d}{dt}(e^{At}) = Ae^{At}$$

$$\left. \frac{d}{dt}(e^{At}) \right|_{t=0} = Ae^{A \cdot 0} = A //$$

$$\frac{d}{dt}(e^{At}) = \begin{bmatrix} -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \\ -4e^{-2t} + 2e^{-t} & -4e^{-2t} + e^{-t} \end{bmatrix}$$

$$\therefore \left. \frac{d}{dt}(e^{At}) \right|_{t=0} = \begin{bmatrix} -2+2 & -1+2 \\ -4+2 & -4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

2020

03

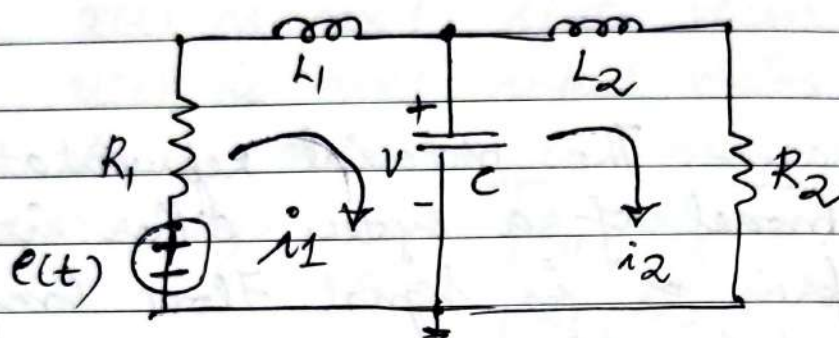
OCTOBER

WK 40 • 27-09
SATURDAY

O	M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11				
12	13	14	15	16	17	18	19	20	21	22	23	24	25	
26	27	28	29	30	31									

STATE-SPACE REPRESENTATION USING PHYSICAL VARIABLES:-

1) Electrical systems



No. of state variables = No. of energy storage elements

= order of the system

Current through the inductor and voltage across the capacitor are taken as state variables

Let x_1 = Current through the inductor $L_1 = i_1$

x_2 = Current through the inductor $L_2 = i_2$

x_3 = Voltage across the capacitor $C = v$

u = Input voltage = $e(t)$

Let voltage across the resistor R_2 be the output of the circuit i.e. $y = i_2 R_2$

04 SUNDAY

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23	24	25	26	27	28	29	30							V

$$e(t) = R_1 i_1 + L_1 \frac{di_1}{dt} + \frac{1}{c} \int (i_1 - i_2) dt \quad - (1)$$

$$0 = L_2 \frac{di_2}{dt} + R_2 i_2 + \frac{1}{c} \int (i_2 - i_1) dt \quad - (2)$$

$$v = \frac{1}{c} \int (i_1 - i_2) dt \quad - (3)$$

From (1) $L_1 \frac{di_1}{dt} = e(t) - R_1 i_1 - \frac{1}{c} \int (i_1 - i_2) dt$

$$\Rightarrow L_1 \dot{x}_1 = u(t) - R_1 x_1 - x_3$$

$$\Rightarrow \dot{x}_1 = \frac{-R_1}{L_1} x_1 - \frac{x_3}{L_1} + \frac{u(t)}{L_1} \quad - (4)$$

From (2) $L_2 \dot{i}_2 = -R_2 i_2 - \frac{1}{c} \int (i_2 - i_1) dt$

$$\Rightarrow \dot{x}_2 = \frac{-R_2}{L_2} x_2 + \frac{x_3}{L_2} \quad - (5)$$

From (3) $c \frac{dv}{dt} = i_1 - i_2$

$$\Rightarrow \frac{dv}{dt} = \frac{i_1}{c} - \frac{i_2}{c}$$

$$\Rightarrow \dot{x}_3 = \frac{x_1}{c} - \frac{x_2}{c} \quad - (6)$$

06

OCTOBER

WK. 41 • 200-198
TUESDAY

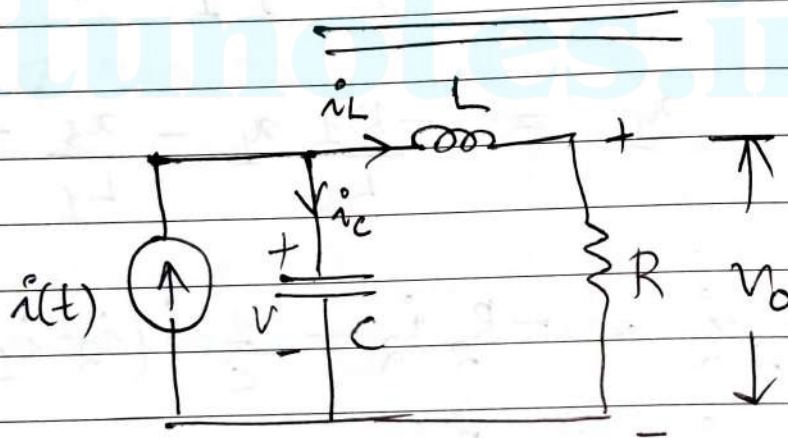
O	T	M	T	F	S	S	M	T	F	S	S
	1	2	3	4	5	6	7	8	9	10	11
12	13	14	15	16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31				

Thus,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{R_1}{L_1} & 0 & -\frac{1}{L_1} \\ 0 & -\frac{R_2}{L_2} & \frac{1}{L_2} \\ \frac{1}{C} & -\frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & R_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

2)



$$x_1 = i_L$$

$$x_2 = v$$

$$u = i(t)$$

$$y = v_o = R i_L$$

$$i(t) = i_L + i_c$$

$$L \frac{di_L}{dt} + R i_L - v = 0$$

$$\Rightarrow \dot{x}_1 = -\frac{R}{L} x_1 + \frac{x_2}{L} \quad \text{--- (1)}$$

$$v = \frac{1}{C} \int i_c dt$$

$$C \frac{dv}{dt} = i_c$$

$$\frac{dv}{dt} = i - i_L$$

2020

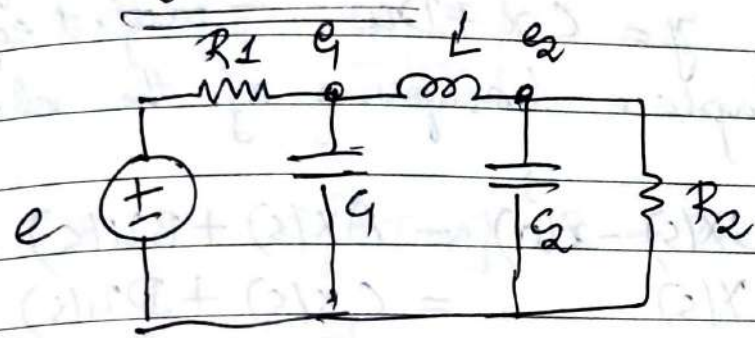
M	T	W	T	F	S	S	M	T	W	T	F	S	S
					1	2	3	4	5	6	7	8	
9	10	11	12	13	14	15	16	17	18	19	20	21	22
23	24	25	26	27	28	29	30						

$$\Rightarrow \frac{dv}{dt} = \frac{i}{C} - \frac{i_2}{C}$$

$$\Rightarrow \dot{x}_2 = \frac{u}{C} - \frac{x_1}{C} \quad - (2)$$

$$y = R x_1 \quad - (3)$$

3)



$$x_1 = e_1$$

$$x_2 = e_2$$

$$x_3 = \frac{1}{L} \int (e_1 - e_2) dt$$

$$0 = C_2 \frac{de_2}{dt} + \frac{e_2}{R_2} + \frac{1}{L} \int (e_2 - e_1) dt$$

$$0 = C_1 \frac{de_1}{dt} + \frac{e_1 - e}{R_1} + \frac{1}{L} \int (e_1 - e_2) dt$$

$$\Rightarrow C_1 \frac{de_1}{dt} = -\frac{e_1}{R_1} + \frac{e}{R_1} - \frac{1}{L} \int (e_1 - e_2) dt$$

$$\Rightarrow \dot{x}_1 = -\frac{x_1}{R_1 C_1} + \frac{u}{R_1 C_1} - \frac{x_3}{C_1} \quad - (1)$$

$$\Rightarrow \dot{x}_2 = \frac{-e_2}{R_2 C_2} + \frac{x_3}{C_2} \Rightarrow \dot{x}_2 = -\frac{x_2}{R_2 C_2} + \frac{x_3}{C_2} \quad - (2)$$

$$\Rightarrow \dot{x}_3 = \frac{e_1}{L} - \frac{e_2}{L} = \frac{x_1}{L} - \frac{x_2}{L} \quad - (3)$$

$$y = \text{voltage across } R_2 = e_2 = x_2 \quad - (4)$$

M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11	12	13	
14	15	16	17	18	19	20	21	22	23	24	25	26	27
28	29	30	31										

2) Mechanical Systems :-

a) $\ddot{y} + 5\dot{y} + 2y = u(t)$

3 state variables x_1, x_2 and x_3

$$x_1 = y \quad \text{--- (1)}$$

$$x_2 = \dot{y} \quad \text{--- (2)}$$

$$x_3 = \ddot{y} \quad \text{--- (3)}$$

$$\dot{x}_1 = \dot{y} = x_2$$

Why $\dot{x}_2 = \ddot{y} = x_3$

$$\begin{aligned} \dot{x}_3 = \dddot{y} &= -5\ddot{y} - 2\dot{y} - y + u(t) \\ &= -5x_3 - 2x_2 - x_1 + u(t) \end{aligned}$$

State model :-

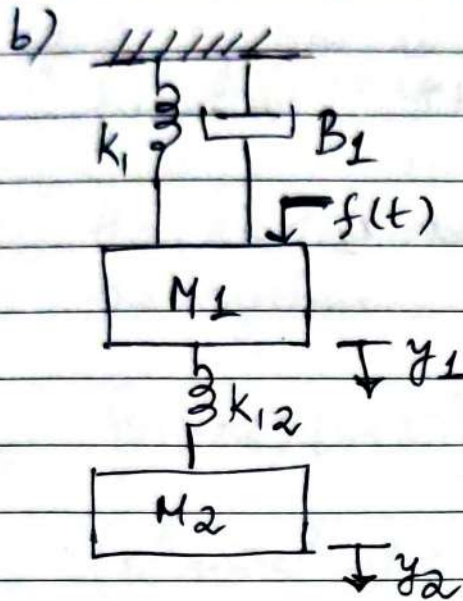
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

05

NOVEMBER

N	M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8							
9	10	11	12	13	14	15	16	17	18	19	20	21	22	
23	24	25	26	27	28	29	30							

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THURSDAY

$$f(t) = M_1 \frac{d^2 y_1}{dt^2} + B_1 \frac{dy_1}{dt} +$$

$$k_1 y_1 + k_{12} (y_1 - y_2)$$

$$\Rightarrow f(t) = M_1 \ddot{y}_1 + B_1 \dot{y}_1 +$$

$$k_1 y_1 + k_{12} y_1 - k_{12} y_2$$

— (1)

$$0 = M_2 \frac{d^2 y_2}{dt^2} + k_{12} (y_2 - y_1)$$

$$\Rightarrow 0 = M_2 \ddot{y}_2 + k_{12} y_2 - k_{12} y_1 \quad \text{--- (2)}$$

Choose state variables

$$x_1 = y_1$$

$$x_2 = \dot{y}_1$$

$$x_3 = y_2$$

$$x_4 = \dot{y}_2$$

State equation:-

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \ddot{y}_1$$

$$= \frac{f(t)}{M_1} - \frac{B_1}{M_1} \dot{y}_1 - \frac{k_1}{M_1} y_1$$

$$- \frac{k_{12}}{M_1} y_1 + \frac{k_{12}}{M_1} y_2 \quad \text{from (1)}$$

$$\Rightarrow \dot{x}_2 = \frac{u(t)}{M_1} - \frac{B_1}{M_1} x_2 - \frac{k_1}{M_1} x_1 - \frac{k_{12}}{M_1} x_1 + \frac{k_{12}}{M_1} x_3$$

2020

M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11	12	13	
14	15	16	17	18	19	20	21	22	23	24	25	26	27
28	29	30	31										

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = \ddot{y}_2 = \frac{-k_{12}}{M_2} y_2 + \frac{k_{12}}{M_2} y_1 \quad \text{from (2)}$$

$$\Rightarrow \dot{x}_4 = \frac{-k_{12}}{M_2} x_3 + \frac{k_{12}}{M_2} x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_1 + k_{12}) & -B_1 & k_{12} & 0 \\ -\frac{M_1}{M_2} & \frac{M_1}{M_2} & 0 & -1 \\ \frac{k_{12}}{M_2} & 0 & -\frac{k_{12}}{M_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/M_1 u(t) \\ 0 \\ 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

M	T	W	T	F	S	S	M	T	W	T	F	S	S
							1	2	3	4	5	6	7
9	10	11	12	13	14	15	16	17	18	19	20	21	22
23	24	25	26	27	28	29	30						

CONCEPTS OF CONTROLLABILITY AND OBSERVABILITY

A system is said to be completely state controllable if it is possible to transfer the system state from any initial state $x(t_0)$ to any desired state $x(t)$ in specified finite time by a control vector $u(t)$.

A system is said to be completely observable, if every state $x(t_0)$ can be completely identified by measurements of the output $y(t)$ over a finite time interval.

KALMAN'S TEST :-

A general n^{th} order multi-input linear-time-invariant system $\dot{x} = Ax + Bu$ is completely controllable if and only if the rank of the composite matrix

$$Q_c = [B \mid AB \mid \dots \mid A^{n-1}B]$$

is n . i.e. $|Q_c| \neq 0$

A general n^{th} order multi-input multi-output linear time-invariant system,

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

is completely observable if and only if the rank of composite matrix

is n. ie $|\phi_0| \neq 0$

(1) Check controllability of the following systems:

12 (a)
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$$

$$Q_c = [B \mid AB]$$

$$= \begin{bmatrix} 2 & -2 \\ 5 & -10 \end{bmatrix}$$

$$|Q_c| = -20 + 10 = -10 \neq 0$$

Hence the system is completely state controllable.

$$(b) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

M	T	W	T	F	S	S	M	T	W	T	F	S	S	N
9	10	11	12	13	14	15	16	17	18	19	20	21	22	O
23	24	25	26	27	28	29	30							V

$$\Phi_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 25 \end{bmatrix}$$

$$|\Phi_c| = 1(-1) = -1 \neq 0.$$

Hence the system is controllable.

2) Check observability of the following systems:

$$(a) \quad \dot{x} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u ; y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

$$\Phi_o = \begin{bmatrix} C^T & A^T C^T \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$|\Phi_o| = -1 \neq 0$$

The system is completely observable.

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OCTOBER

WK 44 • 30-08-1
SATURDAY

$$(b) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 4 & 1 & -2 \\ 1 & 1 & -2 \end{bmatrix}$$

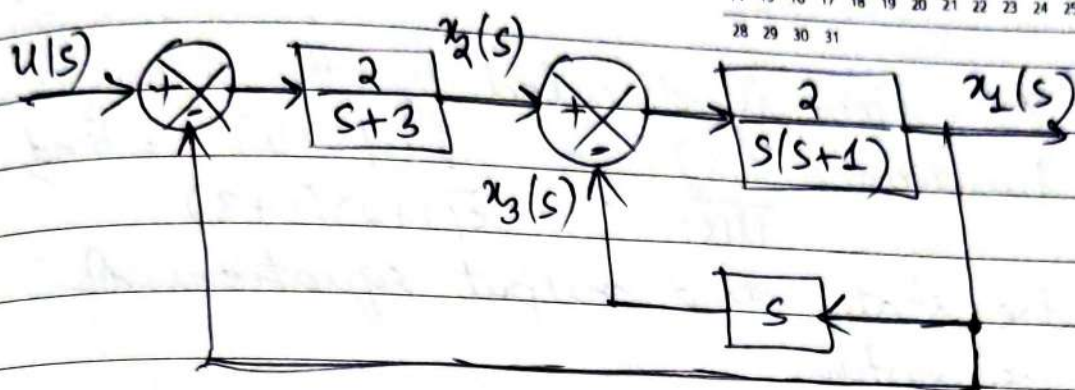
$$|Q_c| = 3(-2+2) = 0$$

The system is not completely observable.

3) Write the state equations of the system shown in fig in which x_1 , x_2 and x_3 constitute the state vector. Determine whether the system is completely controllable and observable.

01 NOV SUNDAY

M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11	12	13	
14	15	16	17	18	19	20	21	22	23	24	25	26	27
28	29	30	31										



$$\left[x_2(s) - x_3(s) \right] \frac{2}{s(s+1)} = x_1(s) \quad - (1)$$

$$x_1(s) s = x_3(s) \quad - (2)$$

$$\left[u(s) - x_1(s) \right] \frac{2}{s+3} = x_2(s) \quad - (3)$$

$$3 \quad (1) \Rightarrow 2x_2 - 2x_3 = s^2 x_1 + s x_1$$

$$4 \quad \Rightarrow 2x_2 - 2x_3 = \ddot{x}_1 + \dot{x}_1 \Rightarrow 2x_2 - 2x_3 = \ddot{x}_1 + \dot{x}_1$$

$$5 \quad (2) \Rightarrow \dot{x}_1 = x_3$$

$$6 \quad (3) \Rightarrow 2u - 2x_1 = \ddot{x}_2 + 3\dot{x}_2$$

$$\text{i.e. } \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u$$

Completely controllable and observable.

2020

Module - 5
State space Analysis

T.F - classical way

$$\frac{L\{o/p\}}{L\{i/p\}}$$

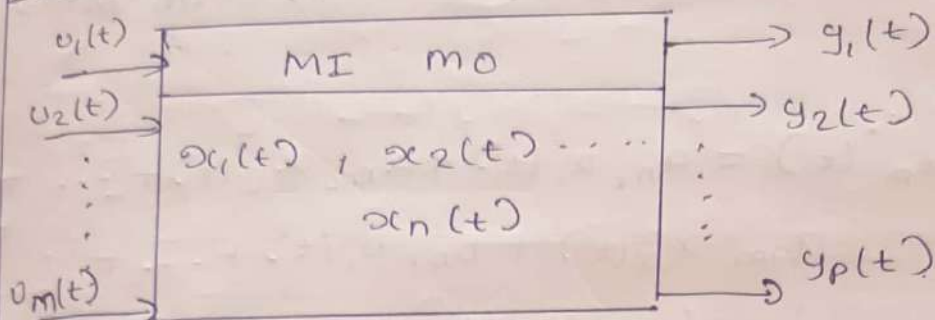
State space analysis - modern way of represent
S/m.

↓
Introduced to overcome difficulties in Transfer
function modelling.

Transfer function	State Space analysis
* Applicable only for Linear Time invariant S/m. * Initial Condition should be set to zero * Applicable only for single i/p Single o/p * Solution of higher order S/m is complex * Represents the relationship b/w o/p & i/p only i.e., no information regarding internal	* Applicable for Linear, Non linear, Time Varying & Time invariant S/m. * No constraints on initial conditions. * Multiple i/p & multiple o/p S/m * Solution is easier since we represent n order S/m as n first order S/m. * Information regarding internal state of S/m is available.

state of the
s/m

* Easier to develop algorithms
in computers



$x_1(t), x_2(t), \dots, x_n(t)$ — State variables

minimum set of variables so that with the knowledge of i/p & state variable we can completely model the s/m.

1. State equation

2. o/p equation

State equation — Set of first order differential eqns based on state variables

$$\frac{dx_1(t)}{dt} = \dot{x}_1(t) = f_1(x_1(t), x_2(t), \dots, x_n(t), u_1(t), u_2(t), \dots, u_m(t))$$

$$\frac{dx_2(t)}{dt} = \dot{x}_2(t) = f_2(x_1(t), x_2(t), \dots, x_n(t), u_1(t), u_2(t), \dots, u_m(t))$$

$$\frac{dx_n(t)}{dt} = \dot{x}_n(t) = f_n(x(t), u(t))$$

Linear

$$\frac{dx_1}{dt} = \dot{x}_1(t) = a_{11}x_1(t) + a_{12}x_2(t) + \dots + a_{1n}x_n(t) + b_{11}u_1(t) + b_{12}u_2(t) + \dots + b_{1m}u_m(t)$$

State equations

$$\begin{cases} \frac{dx_1}{dt} = x_1'(t) = a_{11}x_1(t) + a_{12}x_2(t) + \dots + a_{1n}x_n(t) + b_{11}u_1(t) + \dots + b_{1m}u_m(t) \\ \vdots \\ \frac{dx_n}{dt} = x_n'(t) = a_{n1}x_1(t) + a_{n2}x_2(t) + \dots + a_{nn}x_n(t) + b_{n1}u_1(t) + \dots + b_{nm}u_m(t) \end{cases}$$

matrix form

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}$$

$$\dot{X}(t) = AX(t) + Bu(t)$$

o/p eqn

$$y_1(t) = c_{11}x_1(t) + c_{12}x_2(t) + \dots + c_{1n}x_n(t) + d_{11}u_1(t) + d_{12}u_2(t) + \dots + d_{1m}u_m(t)$$

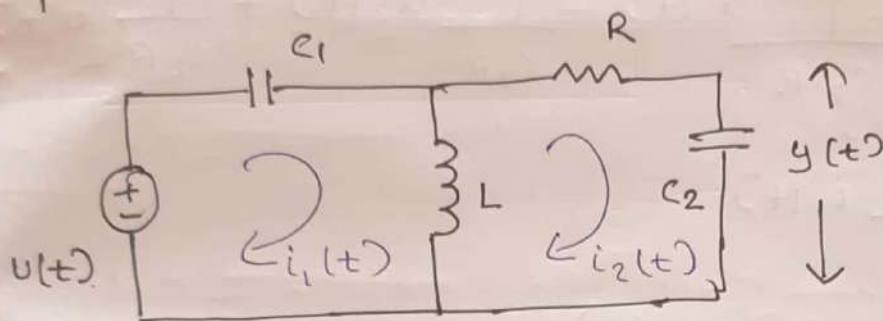
$$y_2(t) = c_{21}x_1(t) + c_{22}x_2(t) + \dots + c_{2n}x_n(t) + d_{21}u_1(t) + d_{22}u_2(t) + \dots + d_{2m}u_m(t)$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{p1} & c_{p2} & \dots & c_{pn} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} + \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1m} \\ d_{21} & d_{22} & \dots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{p1} & d_{p2} & \dots & d_{pm} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}$$

$$Y(t) = CX(t) + Du(t)$$

$$U_m(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}$$

Q) Convert the given circuit into its state space representation



voltage across capacitors } state variables
current through inductor

$$v_{c1} = x_1(t) \quad i_L = x_3(t) = i_1(t) - i_2(t)$$

$$v_{c2} = x_2(t)$$

3 state variables

$$i_{c1} = i_1(t) = c_1 \frac{dv_{c1}}{dt} = c_1 \frac{dx_1}{dt} = c_1 x_1'(t)$$

$$i_{c2} = i_2(t) = c_2 \frac{dv_{c2}}{dt} = c_2 \frac{dx_2}{dt} = c_2 x_2'(t)$$

$$\begin{aligned} v_L &= L \frac{di_L}{dt} = L \frac{d(x_3(t))}{dt} = L \frac{d(i_1(t) - i_2(t))}{dt} \\ &= L \cdot x_3'(t) \end{aligned}$$

KVL,

$$u(t) = v_{c1} + v_L$$

$$u(t) = x_1(t) + L \cdot x_3'(t)$$

$$x_3'(t) = -\frac{1}{L} x_1(t) + \frac{1}{L} u(t) \quad \text{--- ①}$$

loop 2

$$0 = V_2 + V_{C2} + V_L$$

$$= i_2 R + x_2(t) + L \cdot x_3'(t)$$

$$0 = C_2 R \cdot x_2'(t) + x_2(t) - L \left\{ -\frac{1}{L} x_1(t) + \frac{1}{L} u(t) \right\}$$

$$x_2'(t) = -\frac{1}{C_2 R} x_1(t) - \frac{1}{C_2 R} x_2(t) + \frac{1}{C_2 R} u(t) \quad \text{--- (2)}$$

$$x_1'(t) = \frac{1}{C_1} i_1(t)$$

$$i_1(t) = x_3(t) + i_2(t)$$

$$= x_3(t) + C_2 x_2'(t)$$

$$x_1'(t) = \frac{1}{C_1} [x_3(t) + C_2 x_2'(t)]$$

$$= \frac{1}{C_1} \left[x_3(t) + C_2 \left(-\frac{1}{C_2 R} x_1(t) - \frac{1}{C_2 R} x_2(t) + \frac{1}{C_2 R} u(t) \right) \right]$$

$$= \frac{1}{C_1} \left[x_3(t) + -\frac{x_1(t)}{R} - \frac{x_2(t)}{R} + \frac{u(t)}{R} \right]$$

$$= \frac{1}{C_1} x_3(t) + \frac{1}{C_1 R} \left[-x_1(t) - x_2(t) + u(t) \right]$$

$$= \frac{1}{C_1} x_3(t) - \frac{x_1(t)}{R C_1} - \frac{x_2(t)}{R C_1} + \frac{u(t)}{R C_1}$$

$$= -\frac{1}{R C_1} x_1(t) - \frac{x_2(t)}{R C_1} + \frac{1}{C_1} x_3(t) + \frac{u(t)}{R C_1} \quad \text{--- (3)}$$

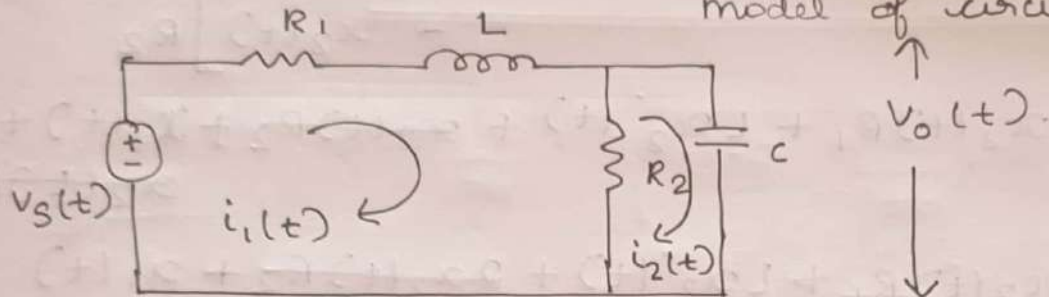
State eqn

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R C_1} & -\frac{1}{R C_1} & \frac{1}{C_1} \\ -\frac{1}{R C_2} & -\frac{1}{R C_2} & 0 \\ -\frac{1}{L} & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R C_1} \\ \frac{1}{R C_2} \\ \frac{1}{L} \end{bmatrix} u(t)$$

$$y(t) = v_{c2} = x_2(t)$$

$$y(t) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} \quad \text{o/p eqn.}$$

③ Select suitable state variables and obtain the model of circuit.



$$v_c = x_1(t)$$

$$i_L(t) = i_R(t) = x_2(t)$$

$$i_c(t) = i_2(t) = C \frac{dv_c}{dt} = C \frac{dx_1(t)}{dt} = C x_1'(t)$$

$$v_L = L \frac{di_L}{dt}$$

$$= L \cdot x_2'(t)$$

$$v_s(t) = v_{R_1} + v_L + v_{R_2}$$

$$= i_1(t) R_1 + L \cdot x_2'(t) + (i_1(t) - i_2(t)) R_2$$

$$= x_2(t) R_1 + L \cdot x_2'(t) + [x_2(t) - C x_1'(t)] R_2$$

Apply KVL in 2nd loop,

$$\cancel{v_o(t)} = \cancel{R_2 i_2(t)} + \cancel{v_c(t)} \\ = \cancel{R_2 i_2(t)} +$$

$$0 = v_c + v_{R_2}$$

$$= x_1(t) + (i_2(t) - i_1(t)) R_2$$

$$= x_1(t) + R_2 [C \cdot x_1'(t) - x_2(t)]$$

$$\dot{x}_1(t) = \frac{-1}{R_2 C} x_1(t) + \frac{1}{C} x_2(t) \quad \text{--- (1)}$$

Sub in (2)

$$\begin{aligned} V_s(t) &= x_2(t) R_1 + L \dot{x}_2(t) + \left[x_2(t) + \frac{1}{R_2} x_1(t) - x_2(t) \right] R_2 \\ &= x_2(t) R_1 + L \dot{x}_2(t) + \cancel{x_2(t) R_2} + x_1(t) + \cancel{x_2(t) R_2} \end{aligned}$$

$$V_s(t) = x_2(t) R_1 + L \dot{x}_2(t) + 2x_2(t) R_2 + x_1(t)$$

$$\dot{x}_2(t) = \frac{1}{L} \left[V_s(t) - x_2(t) R_1 - x_1(t) \right]$$

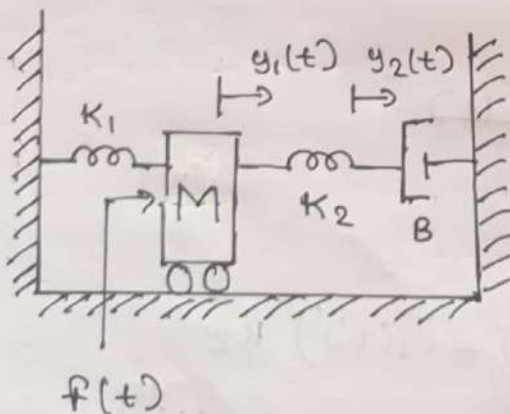
Matrix form

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1/R_2 C & 1/C \\ -1/L & -R_1/L \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} V_s(t)$$

$$y(t) = V_o(t) = V_c = x_1(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Q)



Convert this to
State model ?

← "Mechanical model"

$$m \frac{d^2 y_1}{dt^2} + K_1 y_1(t) + K_2 (y_1(t) - y_2(t)) = f(t) \quad \text{--- (1)}$$

@ node 1

$$B \frac{dy_2}{dt} + K_2 (y_2(t) - y_1(t)) = 0 \quad \text{--- (2)}$$

@ node 2.

$$x_1(t) = y_1(t)$$

$$x_2(t) = y_2(t)$$

$$\frac{dy_1}{dt} = v_1(t) = x_3(t) \quad \left(\begin{array}{l} \text{2nd order ayend } \frac{dy_1}{dt} \text{ en olla} \\ \text{ora term} \\ \text{varanam} \end{array} \right)$$

$$m x_3'(t) + K_1 x_1(t) + K_2 (x_1(t) - x_2(t)) = f(t)$$

$$x_3'(t) = \frac{f(t) - K_1 x_1(t) - K_2 x_1(t) + K_2 x_2(t)}{m}$$

$$= - \frac{(K_1 + K_2)}{m} x_1(t) + \frac{K_2}{m} x_2(t) + \frac{1}{m} f(t)$$

$$\textcircled{2} \Rightarrow B \cdot x_2'(t) + K_2 [x_2(t) - x_1(t)] = 0$$

$$x_2'(t) = \frac{K_2}{B} x_1(t) - \frac{K_2}{B} x_2(t)$$

$$x_1'(t) = \frac{dy_1}{dt} = x_3(t)$$

matrix form

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ K_2/B & -K_2/B & 0 \\ -\frac{(K_1 + K_2)}{m} & \frac{K_2}{m} & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/m \end{bmatrix} f(t)$$

$$y_1(t) = x_1(t)$$

$$y_2(t) = x_2(t)$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

$$\frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_1 \frac{dy}{dt} + a_0 y(t) = b_0 u(t) \quad \text{--- ①}$$

$$y(t) \text{ --- i/p}$$

$$u(t) \text{ --- o/p}$$

$$x_1(t) = y(t)$$

$$x_2(t) = \frac{dy}{dt} = x_1'(t)$$

$$x_3(t) = \frac{d^2 y}{dt^2} = x_2'(t)$$

$$x_n(t) = \frac{d^{n-1} y}{dt^{n-1}}$$

$$\text{①} \Rightarrow x_n'(t) + a_{n-1} x_n(t) + \dots + a_1 x_2(t) + a_0 x_1(t) = b_0 u(t)$$

$$x_n'(t) = -a_{n-1} x_n(t) - \dots - a_1 x_2(t) - a_0 x_1(t) + b_0 u(t)$$

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_0 & -a_1 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u(t)$$

$$y(t) = x_1(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

Q) obtain the state model for the transfer function

$$\frac{Y(s)}{U(s)} = \frac{10}{s^3 + 4s^2 + 2s + 1}$$

$$Y(s)s^3 + Y(s)4s^2 + Y(s)2s + Y(s) = 10U(s)$$

$$\frac{d^3 y(t)}{dt^3} + 4 \frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = 10u(t)$$

$$\text{Let } x_1(t) = y(t)$$

$$x_2(t) = \frac{d}{dt} y(t) = x_1'(t)$$

$$x_3(t) = \frac{d^2}{dt^2} y(t) = x_2'(t)$$

$$x_3'(t) = -x_1(t) - 2x_2(t) - 4x_3(t) + 10u(t)$$

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

Q) $\frac{Y(s)}{U(s)} = \frac{10(s+4)}{s(s+1)(s+3)}$

$$Y(s)s(s+1)(s+3) = U(s)10(s+4)$$

$$Y(s)[s^2 + s(s+3)] = 10sU(s) + 40U(s)$$

$$Y(s)[s^3 + 3s^2 + s^2 + 3s] = 10sU(s) + 40U(s)$$

$$Y(s) [s^3 + 4s^2 + 3s] = 10sU(s) + 40U(s) = 10(s+4)U(s)$$

$$\frac{d^3 y(t)}{dt^3} + 4 \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} = 10 \frac{du(t)}{dt} + 40u(t)$$

$$\frac{K(s)}{U(s)} \cdot \frac{Y(s)}{K(s)} = \frac{10(s+4)}{s(s+1)(s+3)}$$

$$\frac{K(s)}{U(s)} = \frac{1}{s(s+1)(s+3)}$$

$$K(s) [s(s+1)(s+3)] = U(s)$$

$$\frac{d^3 k}{dt^3} + 4 \frac{d^2 k}{dt^2} + 3 \frac{dk}{dt} = U(t)$$

$$x_1(t) = k(t)$$

$$x_2(t) = \frac{dk}{dt} = x_1'(t)$$

$$x_3(t) = \frac{d^2 k}{dt^2} = x_2'(t)$$

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} U(t)$$

$$\frac{Y(s)}{K(s)} = 10(s+4)$$

$$y(t) = 10 \frac{dk}{dt} + 40k(t)$$

$$y(t) = 10x_2(t) + 40x_1(t)$$

$$y(t) = \begin{bmatrix} 40 & 10 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$